

Time Reporting System

System Specification - Course Version

Date: January 2026

Background and System Goals

The system is intended to provide a simple and efficient solution for managing and reporting work hours. It will allow each employee to report hours in a simple, accurate, and organized way, while maintaining transparency between the employee and the system administrator.

Main Goals

- **Improving the reporting process:** Creating a structured and uniform mechanism for reporting hours, reducing errors and data loss.
- **Transparency and control:** Management and tracking with immediate indication of reporting status.
- **Streamlining processes:** Managing vacations and absences in a simple and organized way.
- **Implementing an organized reporting culture:** Establishing a uniform standard for all employees who report hours according to projects and tasks.

Technical Overview

Technologies

- Frontend: React
- Backend: Node.js
- Database: SQL
- עיצוב: Mobile-first responsive
- שפה: Hebrew (RTL) only

Platform

A responsive website accessible from modern browsers (Chrome, Edge, Safari) on desktop and mobile. Mobile adaptation should be in the reporting system; the admin panel is WEB only.

Permissions and User Types

Regular User (Employee)

- Reporting work hours and absences
- Viewing their own reporting history
- Editing reports (until the month is closed)

Admin

- All capabilities of a regular user
 - User management (add, edit, delete)
 - Managing clients, projects, and tasks
 - Assigning users to tasks
 - Editing employee reports
 - Locking the system for reports (month closing)
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Login to the System

- Login using email and password.
 - Users are created by the admin only (including setting an initial password).
 - Saving login details for user convenience – according to information security requirements.
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Daily Hours Report

Default View

When entering the system, the user sees the daily hours reporting input screen. The goal is to make it easier for the user and simplify the process with maximum information and minimum clicks.

On the daily report screen, the following will be displayed:

- The daily standard hours (fixed 9 hours)
 - An indication of reporting progress in relation to the standard hours
 - Existing reports for the same day (if there are any)
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Report Fields

All fields are mandatory, with an indication to the user if a field is missing.

תאריך דיווח

- Date picker field
- Default: current date
- Option to manually select another date

מיקום עבודה

Selection from a list:

- משרד
- לקוח
- בית

שעת התחלה

- Time field
- Default: current time
- Option for manual input
- Option to activate a timer ("התחל עבודה" button)

שעת סיום

- Time field
- Default: current time
- Option for manual input
- Option to stop the timer ("סיום עבודה" button)
- The system automatically calculates the duration

בחירת לקוח

- Dropdown field
- Displays only clients in which the user has assigned tasks
- If there is only one client - it will be displayed automatically
- Client display according to alphabetical order
- The user will have the ability to sort according to reporting frequency

בחירת פרויקט

- Dropdown field
- According to the selected client, the user will be able to choose from the projects in which they have assigned tasks
- If there is only one project - it will be displayed automatically

- Project display according to alphabetical order
- The user will have the ability to sort according to reporting frequency

בחירת משימה

- Dropdown field
- According to the selected project, the user will be able to choose from the tasks they are assigned to
- If there is only one task - it will be displayed automatically
- Task display according to alphabetical order
- The user will have the ability to sort according to reporting frequency

פירוט

- Free text field
 - Description of the work that was done
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Validations

- If שעת התחלה < שעת סיום - an error will be displayed
 - If fewer hours than the daily standard were entered (9 hours) - an alert will be displayed
 - If more hours than the daily standard were entered - an alert will be displayed
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Reporting Process with Timer

Start of Work Day

1. The user enters the system and clicks the “התחלה” button
2. The system starts counting the work time
3. This state is displayed prominently to the user

End of Work Day

1. The user clicks the “סיום” button
2. A window opens to complete the report details:
 - a. בחירת מיקום עבודה
 - b. בחירת לקוח
 - c. בחירת פרויקט

d. בחירת משימה

e. הוספת פירוט

Multiple Task Reporting

- The user can report several clients and tasks on the same day
 - For each row in the report: client, project, task, work hours, and description
 - The system displays how much time remains to be reported out of the total hours
 - The user cannot close the report before all hours have been allocated
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Saving a Report

- After filling in all fields, the user saves the report
 - Indication that the report was saved successfully
 - Option to cancel the report
 - After saving, the screen resets for a new report
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Editing a Report

- Reports can be edited retroactively until the month is closed by the admin
 - After the month is closed, reports are locked for editing
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Absence Reporting

Absence Types (Fixed in the System)

- חופשה
 - מחלה
 - מילואים
 - אחר
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Absence Report Fields

סוג היעדרות

- Dropdown field
- Selection from the fixed list of types

תאריכים

- Option to select a single date
- Option to select a date range (from date to date)
- The system automatically excludes Fridays-Saturdays from the calculation

היעדרות חלקית

- Option to mark partial absence (half day)
- In case of partial absence, a work hours report must be completed for the rest of the day

צירוף מסמכים

- מחלה: mandatory to attach a sick leave certificate
- מילואים: mandatory to attach a reserve duty certificate
- Option to add a document even after the report has been closed (because sometimes the certificate is received later)

Monthly View

The user can view their reports on a monthly basis.

Displayed Information

- Monthly calendar with an indication for each day
- Status of each day: full / missing / exceptional
- Detailed list of reports

Details of Each Report

- תאריך
- משך עבודה (from hour to hour)
- לקוח
- פרויקט
- משימה
- פירוט

Editing Reports

- A report can be clicked and edited
 - Editing is possible retroactively until the month is closed by the admin
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Management Interface (Admin)

User Management

Adding a New User

- שם מלא
- כתובת מייל (used for login)
- סיסמה ראשונית
- סוג משתמש (regular / admin)

Editing a User

- Updating details
- Resetting password
- Changing user type
- Classifying a user as active / inactive

Deleting a User

- Marking as inactive (Soft Delete) - historical data is preserved in the system
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Client Management

Adding a New Client

- שם לקוח
- פרטי קשר (optional)
- Classifying a client as active / inactive – default is that the client is active

Editing a Client

- Updating details
- Changing active / inactive status

Deleting a Client

- Marking as inactive (Soft Delete) - historical data is preserved in the system
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Project Management

Adding a New Project

- בחירת לקוח – according to the list of active clients
- שם פרויקט
- סטטוס: active / inactive – default is active

Editing a Project

- Updating details
- Changing active / inactive status

Deleting a Project

- Marking as inactive (Soft Delete) - historical data is preserved in the system
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Task Management

Adding a New Task

- שיוך לפרויקט – option to add a new task through the project screen
- שם – task description
- סטטוס: open / closed

Editing a Task

- Updating details
- Changing status

Deleting a Task

- Marking as closed (Soft Delete) - historical data is preserved in the system
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Assigning Users to Tasks

- The admin assigns users directly to tasks (not to clients or projects)
 - The user will see in their report only the tasks they are assigned to
 - When selecting a client/project in the report - only those in which the user has assigned tasks will be displayed
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Employee View

A table displaying all employees in the system:

- שם מלא
 - מייל
 - סוג משתמש
 - סטטוס (active/inactive)
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Editing Employee Reports

The admin can:

- View reports of every employee
 - Edit employee reports
 - Every change is documented in the system
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Locking the System for Reports (Month Closing)

In order to maintain a proper reporting process, the admin will have the ability to lock the system for reports:

- Locking a specific month for reports and edits
 - All hour reports of all users in the relevant month are locked – there is no longer an option to edit
 - The system will save the lock timestamp + who performed it
 - The admin can reopen a locked month if needed
 - Any retroactive change after locking requires reopening by the admin
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System Navigation

Regular User

- Hours report screen (Home)
- Monthly view
- Report history

Admin

- All screens of a regular user
- User management
- Client management
- Project management

- Task management
 - Assignments
 - Month closing
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Roadmap - Feature List

The features are arranged according to the recommended work order. Tests must be written for every feature during development.

Setup & Infrastructure

1. Setting up Docker environment (Frontend + Backend + DB)
2. Basic DB structure
3. CI Pipeline
4. CD Pipeline

Authentication

5. User registration (by admin)
6. Login
7. Sessions/JWT management

Admin - Entity Management

8. User management (CRUD)
9. Client management (CRUD)
10. Project management (CRUD)
11. Task management (CRUD)
12. Assigning users to tasks

Hours Reporting - Employee

13. Manual hours reporting
14. Multiple task reporting
15. Editing a report
16. Monthly view

Absences

- 17. Absence reporting
- 18. Attaching documents
- 19. Partial absence

Advanced Features

- 20. Timer (start/end work)
- 21. Month closing (admin)
- 22. Editing employee reports (admin)

Documentation

- 23. Swagger Documentation
 - 24. README
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Technical Requirements

Development and Runtime Environment

רכיב	דרישה
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Containerization Docker + Docker Compose - all services run in containers

Database	PostgreSQL
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Frontend	React
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Backend	Node.js
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CI/CD

רכיב	דרישה
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CI Pipeline GitHub Actions

CD Pipeline Free service of the team's choice (Vercel, Render, Railway, etc.)

CI-טסטים ב All tests must pass before merge

Testing

דרישה רכיב

Unit Tests סוג טסטים

Minimum 60% כיסוי קוד

Team's choice (Jest, Vitest, etc.) כלי בדיקות

Git Workflow

דרישה רכיב

Branch Protection Mandatory on main/master

Pull Requests Mandatory - no direct push to main

Code Review Mandatory before merge

Reviewers 1 מינימום

Documentation

דרישה רכיב

Full installation and run instructions README

API Documentation Swagger

Appendix: System Constants

ערך פרמטר

9 hours תקן שעות יומי

חופשה, מחלה, מילואים, אחר סוגי היעדרויות

משרד, לקוח, בית סוגי מיקום עבודה

רגיל, אדמין סוגי משתמשים